

ARTEMIS FORECASTING

A Personal Geopolitical & Sanctions Forecasting System

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OVERVIEW

Artemis Forecasting is a self-built forecasting system designed to bring analytical discipline to geopolitical and sanctions-related prediction. The project was motivated by a simple problem: making a forecast is easy, but knowing whether that forecast was actually good requires structured tracking, scoring and calibration over time. Artemis addresses this by combining forecast logging, automated accuracy scoring, calibration analysis and direct benchmarking against Good Judgment Open, a public forecasting tournament used by thousands of volunteer forecasters worldwide.

The system was built independently from the ground up, evolving from a simple command-line tool into a full-stack web application with an analytics dashboard, professional visual branding and one-click desktop deployment.

FORECASTING METHODOLOGY

The analytical core of the project — and the part most relevant to political risk work — is its scoring and calibration engine:

- **Brier scoring:** each resolved forecast is scored using the Brier score (the squared error between predicted probability and actual outcome), the standard method used in professional forecasting to measure accuracy.
- **Calibration analysis:** forecasts are grouped into probability deciles (e.g. all forecasts made at ~70% confidence) and checked against how often those forecasts actually came true — surfacing over- or under-confidence that a raw accuracy score would hide.
- **Community benchmarking:** the system imports questions and consensus data from Good Judgment Open, allowing individual forecasts to be scored directly against community and expert medians rather than judged in isolation.
- **Multi-forecaster leaderboard:** performance can be tracked and compared across multiple forecasters, ranked by win rate and Brier score.

SYSTEM & FEATURES

Area	What it does
Forecast management	Log forecasts with probability, rationale and forecaster name; update before resolution; prevent duplicates.
Analytics engine	Computes Brier score, win rate and decile-based calibration; renders results as interactive Chart.js visualisations.
Good Judgment Open integration	Bulk-imports 34+ live GJO questions across economics, markets and geopolitics; batch-forecasting tool; tracks community consensus for direct comparison.
Dashboard	Responsive web dashboard with a leaderboard, calibration chart, and organised active/resolved forecast views.
Data management	CSV export for external analysis; timestamped backups; non-destructive database schema migrations.
Deployment	Packaged as a one-click desktop application (custom icon, Windows shortcut) despite being fully web-based.

TECHNICAL BUILD

Built independently using a Python (SQLite) backend and a responsive HTML/CSS/JavaScript frontend, chosen for zero-configuration deployment and full portability of the underlying database. The system evolved through six development phases — from a minimal three-table, three-script foundation, through an analytics and calibration engine, Good Judgment Open integration, professional visual branding (custom logo, Economist-inspired design system), and finally a packaged desktop application with automated backups. Development was iterative throughout: features were added only in response to real usability problems encountered during use, rather than built to a fixed upfront specification — for example, batch-forecasting tools were added once manual one-at-a-time entry proved too slow for working through 10–15 Good Judgment Open questions in a sitting.

SKILLS DEMONSTRATED

- **Quantitative analysis:** applied statistical scoring (Brier score), calibration analysis and benchmarking methodology drawn from professional forecasting practice.
- **Independent project delivery:** conceived, built and iterated on a full-stack application from zero infrastructure through to a packaged, deployable product.
- **Technical range:** Python and database design on the backend; responsive HTML/CSS/JavaScript and data visualisation (Chart.js) on the frontend.
- **Problem-solving under real constraints:** diagnosed and resolved practical deployment issues (data portability, schema evolution) through iterative refinement.

CONCLUSION

Artemis Forecasting demonstrates a systematic, evidence-based approach to prediction — the same discipline of probabilistic reasoning, calibration and benchmarking against outside views that underpins professional political risk analysis. Beyond the technical build, the project reflects an ongoing personal practice of making and scoring real geopolitical and sanctions-related forecasts against a live community benchmark, rather than treating forecasting as a one-off exercise.